

*** What are your name, GitHub username, CodeCombat username, email address, and any blog/site links?**

Name: Jayant Jain

Github: [jayant1992](#)

Codecombat: Shadowthrone

*** What is your academic background?**

I'm in the third year of my undergraduate studies. I'm majoring in Mechanical Engineering from Indian Institute of Technology, Roorkee.

*** What project do you want to build for CodeCombat?**

I want to work on upgrading the level editor for CodeCombat. I think user-generated content is a very important aspect of the game, especially for the future, since at the moment, the content is nowhere near what is achievable with the existing functionality and potential of the game.

Crowdsourcing the content generation is not only a scalable and efficient solution, it is also likely to lead to the creation of levels which are more innovative and closer to what the audience wants. I don't really need to mention all the other benefits of user-generated content.

Currently, the level editor is powerful, but difficult to use. A first-timer faces a very high barrier of entry, causing most of them to give up before even trying. I'll work on reducing this barrier significantly, making the process of creating levels more intuitive and easy.

*** Why are you the right person to work on this project?**

I've been a hobbyist web and game developer for a couple of years now, and I'm really passionate about it. I'm an active member of the Software Development Section of my campus, which primarily focuses on web development. I've won 2 prizes for the applications I've developed in the annual exhibition.

I have a strong knowledge of web development frameworks, and the MVC architecture. I've used Javascript, PHP, Python, Node.js quite extensively, apart from MVC frameworks like AngularJS, Limonade, ToroPHP, Flask and game development frameworks like Phaser and Box2D. While some of these may not be relevant to CodeCombat, it has helped me gain an intuitive and superior understanding of web and game development, which has enabled me to pick up Coffeescript and Backbone fairly quickly.

I've also participated in a few hackathons, achieving 2nd position in Syntax Error, an on-campus hackathon conducted by Google and sponsored by Github. I took part in Google Cloud Developer Challenge 2013 in a team of 3, and we were selected for the [final round](#). Here's the [link to the app](#). We've subsequently [open sourced it as well](#). I also try to [work on my own](#) open-source (and a few closed-source) applications as well.

*** Why are you interested in CodeCombat?**

I've been gaming for a long, long time, even before I owned a computer of my own. I've played all kinds of games, but I've always been inclined towards strategy games. I started out with real-time strategy games, and slowly moved on to turn-based and PVP games. Though CodeCombat doesn't really fit into any of those categories, I still find it fun to play. The single-player campaign was a tad easy, but the ladder/multiplayer levels are making things interesting. It's also a pretty versatile game, with the audience probably including experienced programmers, as well as those looking to get acquainted with programming. I feel the quality of strategy games has actually gone down a little, game-play wise, with the advent of web and mobile gaming. CodeCombat is more old-school, except with the programming twist, of course.

As for the project I've picked specifically, I've been an amateur UI/UX enthusiast since reading Steve Krug's Don't Make Me Think. I seriously think UI/UX doesn't get the attention it deserves, and I see it as a critical, albeit subtle component of game design. Every popular game that I've personally liked - Braid, Limbo, Angry Birds, Badland, Cut The Rope, Plants vs Zombies, Plague Inc etc - every one of them have a simple, intuitive and amazing UI design, whereas other games with equally good and possibly better gameplay haven't been as popular just because of a poor UI. This is actually much more relevant for mobile gaming, but it's also pretty important for games on the web - I'm sure a lot of feedback you get about the editor is about how difficult it is to use. The fact that the editor is versatile and powerful never gets noticed, simply because most people don't get to the stage where they actually use it.

And I don't think anyone can not be excited by CodeCombat after reading this - [Mission Statement](#)

*** What have you already contributed to CodeCombat?**

I've mostly contributed to usability improvements in the editor, while also fixing a few minor bugs. I've enabled drag-panning for the editor, which I thought was an absolute necessity, since click-to-pan was rather tedious. I've added a function for automatically panning to the selected thing in the treema, which makes editing existing levels significantly easier. I've also added a search bar for the add things column on the right, which removes the necessity of scrolling through the list to find the thing. This is time-saving for those already familiar with the editor, and makes it easier to find things for the users new to the editor.

*** How do you plan to achieve completion of your project?**

I think the current issues with the level editor stem from these issues -

1. The lack of an intuitive and helpful UI -
 - Not enough information - lack of titles and descriptions for things, component properties and other

editor elements

- Absence of control - no undo/repeat button, hard-to-find delete option.
 - Lack of a universal search - This would be incredibly helpful, though hard to implement
 - Information not presented well - scripts tab. Too many arrows need to be clicked to understand what's going on
 - Huge, non-categorised list of components while changing a thang.
 - With its different tabs, which feel more like a navbar, the editor doesn't feel like a single page, which it is.
 - The editor looks a little ugly, and feels hard to use. A sleek looking editor would automatically feel easier to use.
2. The inherent difficulty and tediousness associated with creating a level from scratch -
- Template levels/thangs could be added so that getting started is easier
 - Random terrain generator. I created AoE 2 maps primarily because the randomiser made it ridiculously easy to get started.
 - Guides! There is very little documentation or help for newbies. The only source of information I found was a video on youtube, and that too isn't linked anywhere on the site.
 - It should be easier to add the same thang multiple times, especially floors.
 - The absence of a clear workflow or direction, though I guess this will become less of a problem with better documentation.
 - A side-by-side chat could be very helpful.
3. Lack of incentive for creating levels -
- The level creator's name should be mentioned somewhere
 - Similar to how we have a leaderboard for Archmages, maybe we could have a page for Architects (or something similar)

I think the broad issues I've mentioned are in decreasing order of priority, and I intend to work upon them in that order. I'm planning to do some usability testing with the editor first, since the only user I've tried it on till now is me myself. I don't have any other commitments during the summer. So I think I can spend around 40 hours a week on CodeCombat.

*** Are you open to working in a team with other students on the project?**

Yeah, that would be completely fine.

*** Please predict a schedule with dates and important milestones/deliverables (preferably in two week**

increments). Keep in mind [Hofstadter's law](#).

May 19 - May 30:

- Do usability testing on first-time users, and experienced users too.
- Draft mockups of an improved UI design, and finalize one.
- Plan out how the design is to be implemented, in terms of code.

June 1 - 20:

Work on the UI aspects, such as

- Add information and descriptions wherever necessary
- Improve the scripts tab
- Organize the components tab
- Implement control - undo, repeat, delete.

June 20 - 27:

Mid-term evaluation

June 27 - July 10:

- Decide on how to implement search from a UI point of view
- Add universal search functionality
- Create random terrain generator

July 10 - 25:

- Add template levels and thangs.
- Include clear directions for first time users.
- Link tutorials, videos and wiki pages, creating them wherever necessary
- Usability testing again to evaluate improvements

July 25 - August 11:

- Add Architect leaderboard.
- Implement other social features for incentivising level creation
- Complete any pending parts and fix bugs

August 11 - 18:

- Refactoring
- Documenting
- Get project ready for evaluation

*** Say you finish your project early or it proves impossible. What else would you be interested in working on?**

I'd be interested in implementing rendering through WebGL. WebGL rendering will help in eliminating the performance issues and the slightly sluggish nature of the game, a concern which is raised quite often.

*** What are your past experiences with contributing to open source?**

I've worked on a few personal projects that have been open-sourced. I haven't really contributed to open-source projects of such a large scale. I have worked on some large-scale closed source projects for the Software Development Section of my campus, described below.

*** What other relevant projects have you worked on previously and what knowledge you gained from working on them?**

StreetBuy - A multiplayer tycoon-style game based on the Monopoly board game, implemented with an actual Google Map instead of a board. It is still under the process of development. The server is a Node server with socket.io being used. On the client side, we are working with AngularJS for the web app, and Maps API and frontend stuff like Bootstrap 3 and HTML 5 Canvas.

Know Your Government - A web application for spreading awareness about politicians, elections and the government in India. Developed for the Google Cloud Developer Challenge 2013, we were selected among the finalists. The app is hosted [here](#).

We used the Python SDK for the Google App Engine, Google's Cloud Datastore service for storing data, andAlchemy API for sentiment analysis of news and social content. The [app is open source](#)

Echo - An ebook search engine for the campus intranet of my college. I used PHP on the backend with Limonade as the MVC framework, Apache Solr for the indexing and searching and Google Books API for fetching metadata for books. It's a private repo on Github, I'd be willing to provide you with access if necessary.

The above projects involve use of common web technologies, like jade, sass, jquery, bootstrap, handlebars etc.

In addition, I have worked on a few small games using Phaser, Box2D and HTML 5 Canvas.

*** What other time commitments, such as school work, another job, planned vacation, etc., will you have between May 19 and August 18?**

I have no commitments from May 19 to July 20. I will have regular college from July 21 onwards, though I still expect to be able to dedicate 25-30 hours a week to the project.